

2. [Maximum mark: 4]

Given that $\frac{dy}{dx} = \cos\left(x - \frac{\pi}{4}\right)$ and $y = 2$ when $x = \frac{3\pi}{4}$, find y in terms of x .

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5. [Maximum mark: 5]

The derivative of the function f is given by $f'(x) = \frac{6x}{x^2 + 1}$.

The graph of $y = f(x)$ passes through the point $(1, 5)$. Find an expression for $f(x)$.

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6. [Maximum mark: 7]

A particle moves in a straight line such that its velocity, $v \text{ ms}^{-1}$, at time t seconds is given by

$$v = \frac{(t^2 + 1)\cos t}{4}, \quad 0 \leq t \leq 3.$$

- Determine when the particle changes its direction of motion. [2]
- Find the times when the particle's acceleration is -1.9 m s^{-2} . [3]
- Find the particle's acceleration when its speed is at its greatest. [2]

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5. [Maximum mark: 5]

A company manufactures metal tubes for bicycle frames. The diameters of the tubes, D mm, are normally distributed with mean 32 and standard deviation σ . The interquartile range of the diameters is 0.28.

Find the value of σ .

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4. [Maximum mark: 6]

A discrete random variable, X , has the following probability distribution:

x	0	1	2	3
$P(X=x)$	0.41	$k - 0.28$	0.46	$0.29 - 2k^2$

- (a) Show that $2k^2 - k + 0.12 = 0$. [1]
- (b) Find the value of k , giving a reason for your answer. [3]
- (c) Hence, find $E(X)$. [2]

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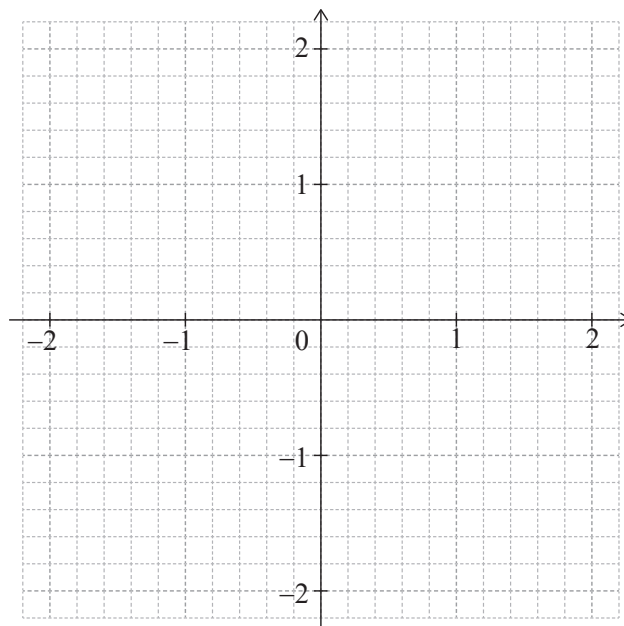


2. [Maximum mark: 5]

Consider the function $f(x) = e^{-x^2} - 0.5$, for $-2 \leq x \leq 2$.

(a) Find the values of x for which $f(x) = 0$. [2]

(b) Sketch the graph of f on the following grid. [3]



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